

Vishal Singh

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PROFESSIONAL SUMMARY

I'm a Computer Science graduate who has spent the last year building real AI systems. I've shipped a multi-agent AI discovery engine, a local RAG assistant, and a computer vision pipeline for autonomous UAVs. My focus is applied GenAI: LLMs, agents, RAG systems, and computer vision — built with Python, LangChain, YOLOv8, and ROS2. I care about making AI that actually works in production, not just in a notebook.

SKILLS

Languages	Python, SQL (PostgreSQL)
AI / ML	PyTorch, Transformers, YOLOv8, Computer Vision, Fine-tuning, Embeddings, Object Tracking
Generative AI	LangChain, RAG, FAISS, Ollama, Llama 3, Mistral, AI Agents, Multi-Agent Systems
Deployment	Netlify, Git, GitHub Actions, Linux (Ubuntu), Supabase, REST APIs
Frontend	Next.js 14, React, Vercel
Platforms	ROS2, AirSim, Salesforce Agentforce, Hugging Face

PROJECTS

Autonomous AI Tool Discovery Engine | Python, Llama 3, Next.js 14, Supabase, GitHub Actions May 2026

- Built a **production-ready multi-agent AI platform** that autonomously discovers, validates, categorises, and continuously updates AI tools using LLM-based verification and Tavily API — fully hands-off.
- Architected a **hallucination-proof pipeline** where Llama 3 cross-references pricing and feature data before any database write, eliminating stale entries.
- Deployed via **GitHub Actions** for zero-maintenance daily operation; served via a Next.js 14 frontend on Vercel.

Live: ai-tool-hunter-eight.vercel.app

GitHub: github.com/VishalsWorkspace/AI-TOOL-HUNTER.git

Autonomous Defence Drone Agent Simulation | Python, ROS2, YOLOv8, AirSim, Ultralytics July 2026

- Developed an **autonomous UAV surveillance pipeline** integrating computer vision, multi-object tracking, threat classification, and decision intelligence for real-time monitoring.
- Fine-tuned **YOLOv8** on custom defence datasets; achieved **15–25 FPS** via a modular perception → tracking → threat classification → alert pipeline within a Gazebo physics simulation.
- Selected for **KSCST State-Level Project Exhibition 2026**.

GitHub: github.com/VishalsWorkspace/aidf-drone-framework.git

LLM Research Assistant (RAG) | LangChain, Ollama, FAISS, Mistral, Sentence Transformers Nov 2025

- Built a **local-first PDF analysis tool** using Retrieval-Augmented Generation (RAG) — private, sub-second document queries with zero cloud dependency.
- Optimised **FAISS vector retrieval** with quantised Mistral models and Sentence Transformer embeddings for low-latency local LLM inference.

Live: llm-pdf-research-assistant.netlify.app/

GitHub: github.com/VishalsWorkspace/LLM_Research_Assistant_2.0.git

EDUCATION

BTech in Computer Science and Engineering — Reva University, Bengaluru July 2026

ACHIEVEMENTS & CERTIFICATIONS

- KSCST State-Level Project Exhibition 2026** — Selected for autonomous UAV surveillance project.
- Salesforce Certified Agentforce Specialist (2025)** — Achieved 100% in Data Cloud integration; validated in autonomous AI agents and enterprise automation workflows.
- Prompt Tuning for NLP — Hugging Face (2025)** — Fine-tuning transformer models for NLP applications.
- IBM SkillsBuild — Generative AI Fundamentals & Prompt Engineering (2024)**.